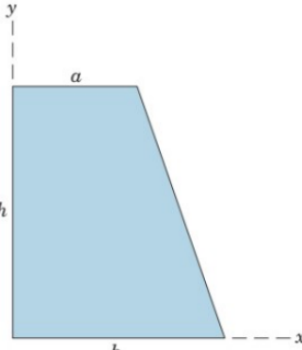
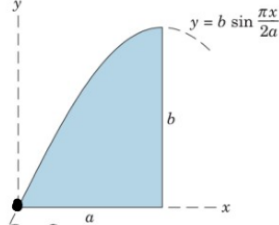


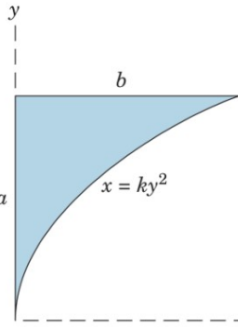
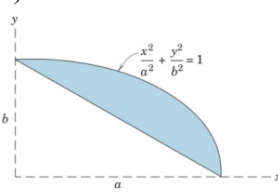


MECHANICS-02

1. Column-I represents uniform density. Column-II represents x-coordinate and Column – III represents y-coordinate of centre of mass of object in column – I w.r.t origin.

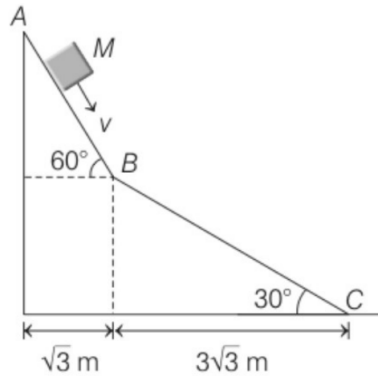
Column-I	Column-II	Column-III
A) Shaded lamina of uniform density 	P) $\frac{2a}{\pi}$	W) $\frac{h(2a+b)}{3(a+b)}$

B) Shaded lamina of uniform density 	Q) $\frac{a}{3\left(\frac{\pi}{2}-1\right)}$	X) $\frac{3a}{4}$
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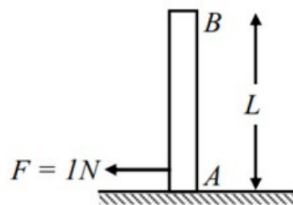
C) Shaded lamina of uniform density 	R) $\frac{a^2+b^2+ab}{3(a+b)}$	Y) $\frac{b}{3\left(\frac{\pi}{2}-1\right)}$
D) Shaded lamina of uniform density 	S) $\frac{3b}{10}$	Z) $\frac{\pi b}{8}$

Paragraph for Question Nos. 2 to 4:

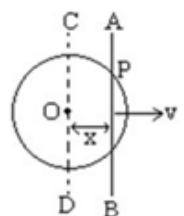
A small block of mass M moves on a frictionless surface of an inclined plane, as shown in figure. The angle of the incline suddenly changes from 60° to 30° at point B. The block is initially at rest at A. Assume that collisions between the block and the incline are totally inelastic ($g = 10 \text{ m/s}^2$).



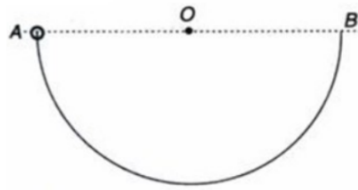
2. The speed of the block at point B immediately after it strikes the second incline is
 A) $\sqrt{60} \text{ m/s}$ B) $\sqrt{45} \text{ m/s}$ C) $\sqrt{30} \text{ m/s}$ D) $\sqrt{15} \text{ m/s}$
3. The speed of the block at point C, immediately before it leaves the second incline is
 A) $\sqrt{120} \text{ m/s}$ B) $\sqrt{105} \text{ m/s}$ C) $\sqrt{90} \text{ m/s}$ D) $\sqrt{75} \text{ m/s}$
4. If collision between the block and the incline is completely elastic, then the vertical (upward) component of the velocity of the block at point B, immediately after it strikes the second incline is
 A) $\sqrt{30} \text{ m/s}$ B) $\sqrt{15} \text{ m/s}$ C) zero D) $-\sqrt{15} \text{ m/s}$
5. A uniform smooth rod of mass $m = 1 \text{ kg}$ and length L is balanced in the vertical position as shown in the figure. When a horizontal force F is applied at end A, the magnitude of acceleration of top point B at this instant is



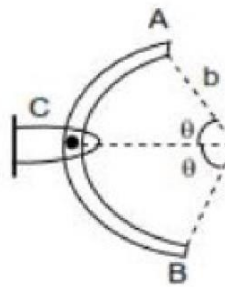
6. A rod AB is moving on a fixed circle of radius $R \text{ m}$ with constant velocity ' v ' as shown in fig. 'P' is the point of intersection of the rod and the circle. At an instant the rod is at a distance $x = \left(\frac{3R}{5}\right) \text{ m}$ from centre of the circle. The velocity of the rod is perpendicular to the rod and the rod is always parallel to the diameter CD. The speed of the point of intersection P in ms^{-1} is nv . Find the value of n .



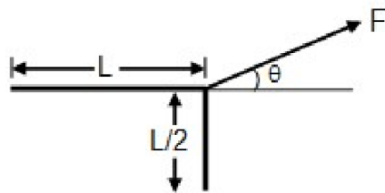
7. A uniform semicircular wire of mass M is hinged at 'A' so that it can rotate freely in vertical plane about a horizontal axis through 'A'. The semicircle is released from rest when its diameter AB is horizontal. $Mg\sqrt{\frac{1}{N^2} + \frac{1}{\pi^2}}$ is the hinge force at 'A' immediately after the wire is released. Find N .



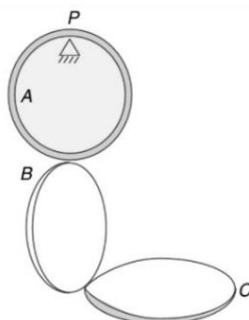
8. A uniform rod AB bent in the shape of an arc of circle is free to rotate in a vertical plane about a fixed horizontal axis passing through the mid – point of the wire. The situation is shown in the figure. The angular acceleration of the rod immediately after it is released from rest from the position shown is $\alpha = \frac{pg}{4b}$. Calculate p .



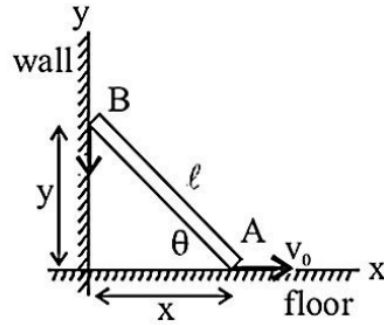
9. A uniform thin L shaped rod is placed on the rough horizontal surface and its corner is pulled by a string as shown in the figure. If it undergoes pure translation then the value of $\cot \theta$ will be



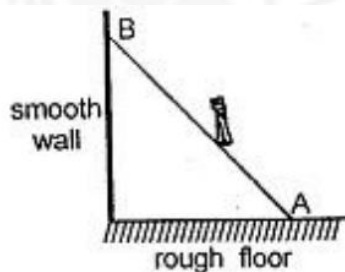
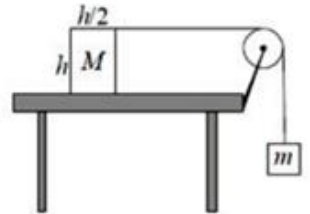
10. Three identical rings each of mass M and radius R are welded together with their planes mutually perpendicular to each other. Ring A is vertical and B is also vertical in a plane perpendicular to A . Ring C is in horizontal plane. The moment of Inertia of this system about a horizontal axis perpendicular to the plane of the figure passing through point P (top point of ring A) is nMR^2 . The value of n is ____



11. A rigid rod is sliding. At some instant, position of the rod is as shown in the figure. End A has constant velocity v_0 along the surface as shown. At $t=0, y=\ell$.



- A) Graph of 'y' as a function of time is a parabola.
- B) component of velocity of end B along the length of the rod is $\frac{v_0 y}{\sqrt{x^2 + y^2}}$
- C) Graph of y as a function of time is an ellipse.
- D) Angular velocity of rod is constant.
12. A table has a heavy circular top of radius 1 m and mass 2 kg. It is supported by four light legs of length 1 m fixed symmetrically on its circumference. Find the maximum mass (in kg) which may be placed anywhere on this table without toppling it. (take $\sqrt{2} = 1.4$)
13. A cylinder of height h, diameter $h/2$ and mass M and with a homogeneous mass distribution is placed on a horizontal table. One end of a string running over a pulley is fastened to the top of the cylinder, a body of mass m is hung from the other end and the system is released. Friction is negligible everywhere. At what minimum ratio m/M will the cylinder tilt?
14. A ladder AB is supported by a smooth vertical wall and rough horizontal floor as shown. A boy starts moving from A to B slowly. The ladder remains at rest, then pick up the correct statements(s):

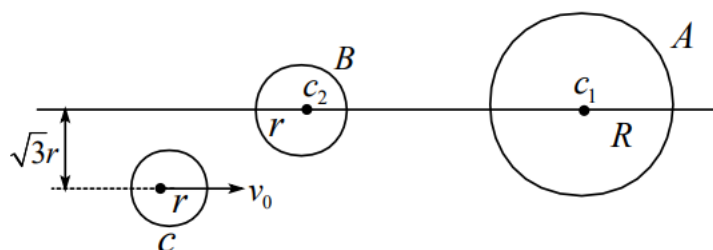


- A) Magnitude of normal reaction by wall on ladder at point B will increase.
- B) Magnitude of normal reaction by wall on ladder at point B will decrease.
- C) Magnitude of normal reaction by floor on ladder at point A will remain unchanged
- D) Magnitude of friction force by floor on ladder at point A will increase.

15. A rigid body is rotating about fixed axis of rotation under the action of different forces acting on it. Choose the correct option(s) from the following
- A) Torque on the body due to a force about axis of rotation is equal to the component of torque due to that force about any point on the axis of rotation along the axis of rotation
- B) Component of torque due to a force along the axis of rotation depends on the point on the axis of rotation about which torque due to that force is considered
- C) Net torque on the body about the axis of rotation is equal to sum of the components of torques due to all forces about any point on the axis of rotation
- D) None of these statements are correct
16. A rigid body is rotating about a fixed axis so that its angular velocity depends on total angle of rotation θ as $\omega = \omega_0 - \alpha\theta$ where ω_0 and α are positive constants and at time $t = 0$, $\theta = 0$. The correct option(s) of the following is/are
- A) θ at any instant is $\frac{\omega_0}{\alpha}(1 - e^{-\alpha t})$ B) θ at any instant is $\frac{\omega_0}{\alpha}e^{-\alpha t}$
- C) Angular velocity at any instant is $\omega_0 e^{-\alpha t}$
- D) Number of rotations made by the body before coming to rest is $\frac{\omega_0}{2\pi\alpha}$

Paragraph for Question Nos. 17 to 18:

Disc "A" of radius R is lying flat on a smooth horizontal surface. Disc B is also at rest. Disc C, which is identical to disc B is traveling along the surface with its velocity (v_0) parallel to the line joining the centres C_1 and C_2 of the discs A & B. The distance between the line C_1C_2 and line of motion of centre of disc C is $(\sqrt{3})r$, where r is the radius of both discs B & C. Impact of C with B is completely elastic and friction is negligible.



17. The ratio of speeds of B & C immediately after their impact is

- A) $\sqrt{3}$ B) 1 C) $\sqrt{2}$ D) $\frac{1}{\sqrt{3}}$

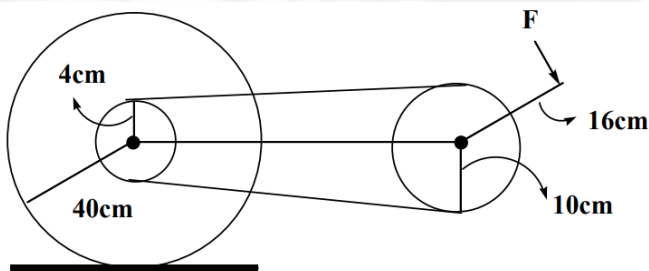
18. The value of radius (R) of disc A such that both discs B & C just miss hitting the disc A subsequently is

- A) $\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)r$ B) $\left(\frac{2}{\sqrt{3}+1}\right)r$ C) $\left(\frac{\sqrt{3}}{\sqrt{3}+1}\right)r$

D) for any value of R, Any one of the discs B,C surely will hit the disc A

Paragraph for Question Nos. 19 to 20:

A bicycle has pedal rods of length 16cm connected to a spoked disc of radius 10cm. The bicycle wheels are 80cm in diameter and the chain runs over a gear of radius 4cm. The cyclist applies a force F perpendicular to the pedal rod, but the cycle is at rest. Assume that tension in the lower part of the chain is negligible and deformation at the point of contact of rear wheel is zero. Neglect any other frictional force within the parts of the bicycle.



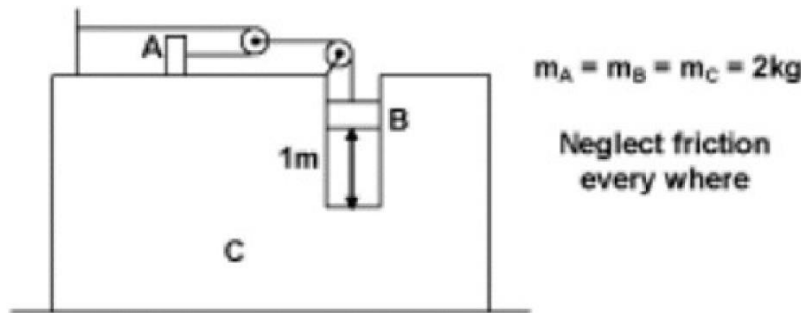
19. If the force applied by the person is $F = 100N$ but the bicycle is at rest, the correct option(s) of the following is/are

- A) Tension in the upper part of the chain is 40N C) Frictional force on the rear wheel is 16N
B) Tension in the upper part of the chain is 160N D) Frictional force on the rear wheel is 12N

20. When the person applies a force $F^{\perp}$ perpendicular to the pedal rod the bicycle is on the verge of moving and frictional force on the front wheel is 20N. The value of $F^{\perp}$ is

- A) 250N B) 175N C) 105N D) 125N

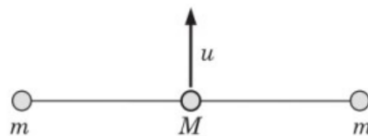
21. Consider the situation given. When system released from rest, block B moves vertically downwards by 1m. The magnitude of displacement of C.O.M. of entire system till the instant block B makes vertical downward displacement of 1m is _____ m.



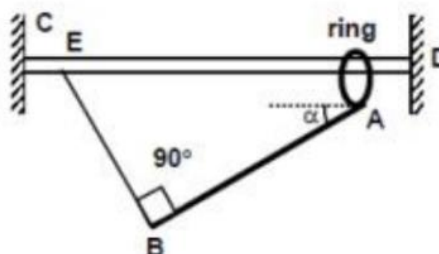
22. A plank of length $l = 2.0$ m rests on a calm water close to a frictionless platform A that is a distance $b = 17$ m from another platform B as shown in the figure. A block of mass $m = 25$ kg sliding on the platform A lands on the plank and stays on it. Force of water resistance on the plank is given by equation $\vec{F} = -k\vec{v}$, where $\vec{F}$ in N is the force of water resistance, $\vec{v}$ in m/s is velocity of the plank and $k = 15$ N.s/m. The minimum speed of the block on the platform A so that the plank reaches the platform B is _____ m/s.



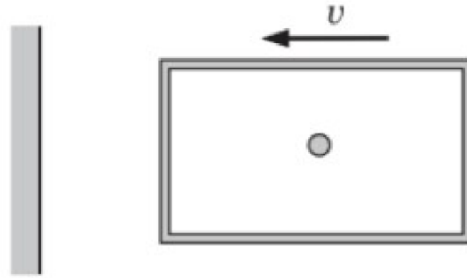
23. A disc of mass $M = 2.0$ kg is connected with two identical discs each of mass $m = 1.0$ kg with the help of two light inextensible threads each of lengths $l = 50$ cm. Initially the system rests on a frictionless horizontal floor with centres of the discs in a straight line and the threads straight. Now the middle disc is projected along the floor with a velocity $u = 2$ m/s perpendicular to the threads. The tensile forces in the threads when the end discs are about to collide is _____ N.



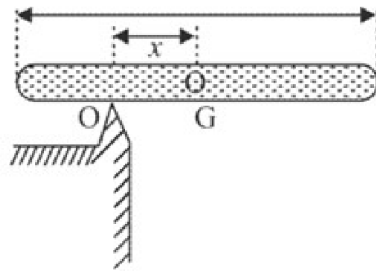
24. One end of a heavy uniform rod AB can slide along rough horizontal guiding surface CD with the help of massless ring as shown in the figure. The dimensions of ring is negligible. BE is the ideal string. If $\angle EBA$ is right angle and α is the angle between rod AB and horizontal when the rod is on the verge of sliding. If the coefficient of friction between ring and horizontal guiding surface CD is μ , then the value of 3μ is _____ (given $\alpha = 45^\circ$)



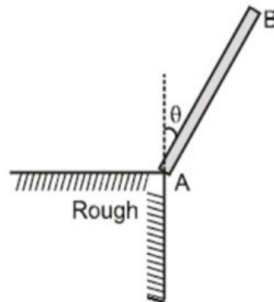
25. In gravity free space, a small ball is placed at the centre of a box of the same mass. The ball and the box both are moving with a velocity v towards a wall. Assume all collisions to be perfectly elastic. Total _____ collisions are possible.



26. In the adjacent figure a uniform rod of length ℓ and mass m is kept at rest in horizontal position on an elevated edge. The value of x (consider the figure) is such that the rod will have maximum angular acceleration α , as soon as it is set free.

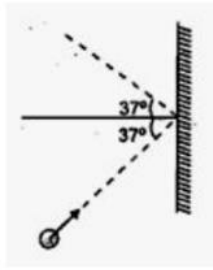


- A) x is equal to $\frac{\ell}{2\sqrt{3}}$ B) α is equal to $\frac{g\sqrt{3}}{2\ell}$
 C) α is equal to $\frac{g\sqrt{3}}{\ell}$ D) x is equal to $\frac{\ell}{\sqrt{3}}$
27. A uniform rod of mass m and (length ℓ is released from rest in the vertical position ($\theta = 0^\circ$) on a rough (surface is sufficiently rough to prevent just sliding) square corner A, as shown in figure, then choose correct options.

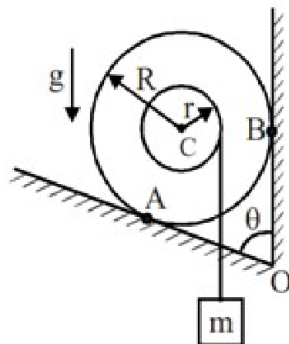


- A) If the rod just begins to slip when $\theta = 37^\circ$ with vertical, then the co-efficient of static friction μ between the rod and the corner is 0.3
- B) If the end of the rod is notched so that it cannot slip, then the angle θ at which contact between the rod and the corner ceases is 53°
- C) If the rod just begins to slip when $\theta = 37^\circ$, then the coefficient of static friction between the rod and the corner is $\frac{3}{4}$
- D) If the end of the rod is notched so that it cannot slip, then the angle θ at which contact between the rod and the corner ceases is 37°

28. A particle moving on a smooth horizontal surface strikes a stationary wall. The angle of strike is equal to the angle of rebound & is equal to 37° and the coefficient of restitution with wall is $e = \frac{1}{5}$. The friction coefficient between wall and the particle in the form $\frac{X}{10}$ and fill value of X.



29. A massless spool of inner radius r , outer radius R is placed against vertical wall and tilted split floor as shown in the figure. A light inextensible thread is tightly wound around the spool through which a mass m is hanging. There exists no friction at point A, while the coefficient of friction between spool and point B is μ . The angle between two surfaces is θ .



A) The magnitude of force on the spool at B in order to maintain equilibrium is

$$mg\sqrt{\left(\frac{r}{R}\right)^2 + \left(1 - \frac{r}{R}\right)^2 \frac{1}{\tan^2 \theta}}$$

B) The magnitude of force on the spool at B in order to maintain equilibrium is

$$mg\left(1 - \frac{r}{R}\right) \frac{1}{\tan \theta}$$

C) The minimum value of μ for the system to remain in equilibrium is $\frac{\cot \theta}{(R/r) - 1}$

D) The minimum value of μ for the system to remain in equilibrium is $\frac{\tan \theta}{(R/r) - 1}$

30. A uniform triangular plate ABC of mass m is placed such that the coordinates of A, B and C are $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$ respectively. Let I_x, I_y and I_z be the moments of inertia of this plate about x-axis, y-axis and z-axis respectively. Then, which of the following are correct.

A) If $a = c$, then $I_x = I_z$

B) I_y is independent of b

C) $I_z = \frac{m}{3}(a^2 + b^2)$

D) If $\frac{a}{5} = \frac{b}{4} = \frac{c}{3}$, then $I_x = \frac{ma^2}{6}$

KEY:

1-A-R-W; B-P-Z; C-S-X; D-Q-Y	2-B	3-B	4-C	5-2	6-1.25
7-2	8-2	9-4	10-29	11-C	12-5
13-1	14-ACD	15-AC	16-ACD	17-D	18-A
19-BC	20-D	21-0.33	22-9	23-2	24-1
25-3	26-AC	27-AB	28-5	29-AD	30-ABD